

Qatar-1b

R-2008

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_220805 · created 2026-08-07T16:49:08+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459796.9403 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.014
Transit depth (Rp/Rs)^2	3.71 +/- 0.52 [%]
Orbital Inclination (inc)	82.96 +/- 1.19
Airmass coefficient 1 (a1)	0.959 +/- 0.018
Airmass coefficient 2 (a2)	0.026 +/- 0.018
Scatter in the residuals of the lightcurve fit is	1.89 %
Best Comparison Star	#2 - [172, 473]
Optimal Aperture	3.47
Optimal Annulus	9.2
Transit Duration (day)	0.0656 +/- 0.0087

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.014 vs 0.1463 (deviation 3.17σ)
Duration fit vs published	0.0656 d vs 0.0692 d (deviation 0.4σ)
Mid-time O–C	-6.0 min
Depth SNR	13.3
Residual scatter	1.89 %

Light curve

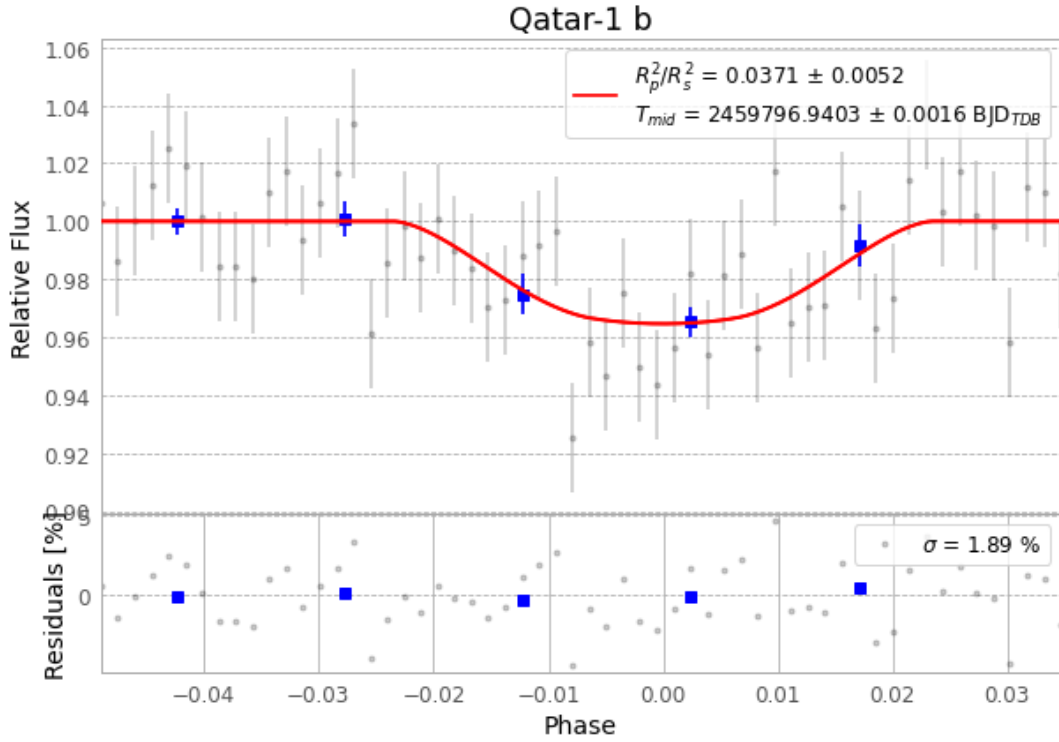


Figure 1 — FinalLightCurve_Qatar-1 b_2022-08-05.png (SHA-256 dabe4255085ec48c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [242, 452], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 1e8e4967ccd96c87...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

59 FITS frames (39 MB), Qatar-1220805085113.FITS → Qatar-1220805114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-220805.log (0b2db200cafe...), FinalLightCurve_Qatar-1 b_2022-08-05.png (dabe4255085e...), FinalLightCurve_Qatar-1 b_2022-08-05.csv (3ef82d2d5b10...)

Compute resources

Wall time 174.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-07 16:49Z.